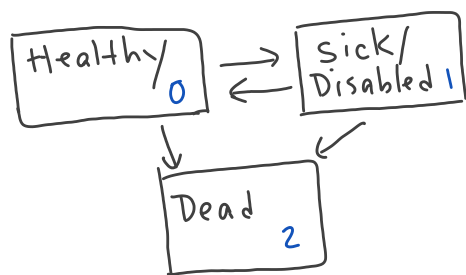
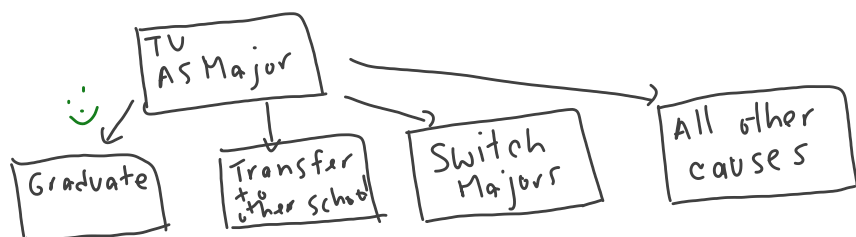


Disability Income Insurance Model



AS Majors



8.3 Assumptions and Notation

• We have a finite set of $(n+1)$ states: $0, 1, 2, \dots, n$ with instantaneous transitions between selected pairs of states.

• For each $t \geq 0$,

$Y(t)$ takes one of the values $0, 1, 2, \dots, n$

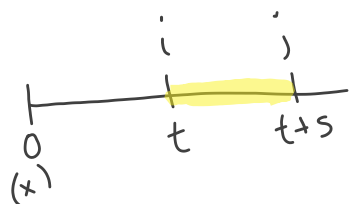
$Y(t) = i$ individual is in state i at age $x+t$

Assumption 1 We assume that for states i and j and times t and $t+s$ where $s \geq 0$,

$P[Y(t+s)=j | Y(t)=i]$ is well defined & does not depend on any information before time t

Given this individual is in state i at time t , what is the probability they will be in state j at time $t+s$

This is called the Markov property.



Assumption 2 $P(2 \text{ or more transitions in an instant of time}) = 0$

${}_tP_x^{ij}$ = Prob that a person age x who is currently in state i will be in state j at time t

i j
 $\begin{array}{|c|c|} \hline 0 & t \\ \hline \end{array}$
 (x)

$= P[Y(x+t)=j | Y(x)=i] \quad i \neq j \text{ OR } i=j$

${}_t\bar{P}_x^{ii}$ = Prob that (x) who is currently in state i stays continuously in state i for t years

$= P[Y(x+s)=i \text{ for all } s \in [0, t] | Y(x)=i]$

Alive-Dead Model



$${}_tP_x^{00} = {}_t\bar{P}_x^{00} = {}_tP_x$$

$${}_tP_x^{01} = {}_tq_x$$

$${}_tP_x^{10} = 0$$

$${}_tP_x^{11} = 1$$

~~No notation like this~~
 ${}_tq_x^{01}$

Matrix

$$\begin{bmatrix} {}_tP_x^{00} & {}_tP_x^{01} & \dots & {}_tP_x^{0n} \\ {}_tP_x^{10} & {}_tP_x^{11} & \dots & {}_tP_x^{1n} \\ \vdots & \vdots & \ddots & \vdots \\ {}_tP_x^{n0} & \dots & \dots & {}_tP_x^{nn} \end{bmatrix} \begin{array}{l} \rightarrow \text{add across } \Sigma = 1 \\ \Sigma = 1 \\ \vdots \\ \Sigma = 1 \end{array}$$

adding down makes no sense

$$\begin{bmatrix} {}_tP_x^{00} & {}_tP_x^{01} \\ {}_tP_x^{10} & {}_tP_x^{11} \end{bmatrix} = \begin{bmatrix} {}_tP_x & {}_tq_x \\ 0 & 1 \end{bmatrix}$$

μ_{x+t} : force of mortality

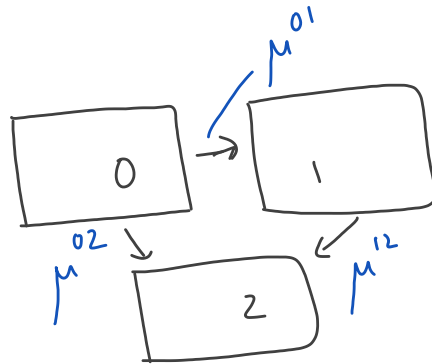
μ_{x+t}^{ij} = force of transition from state i to state j
 $i \neq j$

8.4 Formulas for probabilities

$${}_t p_x = e^{-\int_0^t \mu_{x+s} ds}$$

How does a person stay alive -
 resist the force of mortality
 for t years

$${}_t p_x^{01}$$



$${}_t \bar{p}_x^{00} \quad \begin{matrix} \mu^{01} \\ \mu^{02} \end{matrix}$$

For any state i in a multi-state

$${}_t \bar{p}_x^{ii} = e^{-\int_0^t \left(\sum_{\substack{j=0, \\ j \neq i}}^n \mu_{x+s}^{ij} \right) ds}$$

$${}_t p_x^{ij}$$

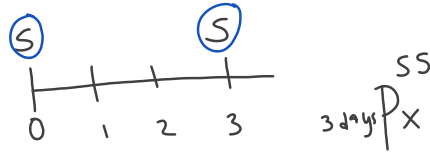
8.14.

Patients are classified as Sick (S), Critical (C), or Discharged (D). Transitions occur daily according to the following Markov transition matrix:

ALTAM
MC

Markov

	S	C	D
S	0.60	0.20	0.20
C	0.10	0.50	0.40
D	0.00	0.00	1.00



Calculate the probability that a patient who is classified as Sick today will be classified as Sick three days later.

- (A) 0.216
(B) 0.234
(C) 0.250
(D) 0.267
(E) 0.284

$$\begin{aligned}
 & S \rightarrow S \rightarrow S \rightarrow S \\
 & (0.6)(0.6)(0.6) = 0.216 \\
 & S \rightarrow C \rightarrow C \rightarrow S : 0.01 \\
 & S \rightarrow S \rightarrow C \rightarrow S : 0.012 \\
 & S \rightarrow C \rightarrow S \rightarrow S : 0.012 \\
 & \hline
 & 0.25
 \end{aligned}$$

[This is a modified version of Question 3 on the Fall 2014 Multiple Choice exam.]

151. For a multi-state model with three states, Healthy (0), Disabled (1), and Dead (2):

- (i) For $k = 0, 1$:

MLC

$$\begin{aligned}
 P_{x+k}^{00} &= 0.70 \\
 P_{x+k}^{01} &= 0.20 \\
 P_{x+k}^{10} &= 0.10 \\
 P_{x+k}^{12} &= 0.25
 \end{aligned}$$

$$\begin{bmatrix}
 0.70 & 0.20 & 0.10 \\
 0.10 & 0.65 & 0.25 \\
 0 & 0 & 1
 \end{bmatrix}$$

P_x

- (ii) There are 100 lives at the start, all Healthy. Their future states are independent.

Calculate the variance of the number of the original 100 lives who die within the first two years.

- (A) 11
(B) 14
(C) 17
(D) 20
(E) 23

$$\begin{aligned}
 P &= {}_2P_x^{02} \\
 & \begin{array}{l}
 0 \rightarrow 0 \rightarrow 2 \quad (0.7)(0.1) \\
 0 \rightarrow 1 \rightarrow 2 \quad (0.2)(0.25) \\
 0 \rightarrow 2 \rightarrow 2 \quad (0.1)(1)
 \end{array} \\
 & \hline
 & 0.22
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}(N) &= 100(0.22)(0.78) \\
 &= 17.16
 \end{aligned}$$

8.8. In a homogeneous Markov model with three states: Healthy (H), Sick (S), and Dead (D), you are given:

(i) The monthly transition probabilities are:

	H	S	D
H	0.75	0.20	0.05
S	0.30	0.50	0.20
D	0.00	0.00	1.00

(ii) Initially there are 10 Healthy lives with independent future states.

Calculate the probability that exactly 4 lives will die during the first two months.

- (A) 0.0005
- (B) 0.0245
- (C) 0.1132
- (D) 0.2136
- (E) 0.4414

$$p(\text{success}) = \frac{3}{12} P_x^{HD} = 0.1275$$

$H \rightarrow H \rightarrow D$
 $H \rightarrow S \rightarrow D$
 $H \rightarrow D \rightarrow D$

Binomial

$$\binom{n}{y} p^y \cdot (1-p)^{n-y}$$

$$\binom{10}{4} (0.1275)^4 (1-0.1275)^6$$

$$= 0.02448$$